

# Roth's Theorem with Random Common Differences

GREEN-010

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

## ORIGINAL PROBLEM STATEMENT

Let  $S \subset \mathbb{N}$  be random. Under what conditions is Roth's theorem for progressions of length 3 true with common differences in  $S$ ?

**Source and attribution.** Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.  
Archive: <https://www.unsolvedmath.com/problems?difficulty=1>  
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>